DriftSense: Intention-Aware Prediction and Model-Assisted Reduction of Browser-Based Digital Drift

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Digital wellbeing tools commonly rely on domain blocking, screen-time limits, and usage dashboards. These approaches can misclassify purposeful browsing as problematic because the same website may support information seeking, intentional breaks, passive consumption, or avoidance. We introduce DriftSense, a privacy-preserving Chrome extension and local machine-learning pipeline for studying browser-based digital drift, defined as self-reported mismatch between browsing intention and session outcome. DriftSense records lightweight session metadata and activity-window counts on configured domains, asks post-session reflection questions, and can optionally ask pre-session intention prompts. The planned study has two stages. First, we collect prompted sessions to train and evaluate drift-risk models against time-threshold, domain-category, intention-only, and activity-only baselines. Second, we freeze the trained model and compare three field-study conditions: passive monitoring, static intention prompting, and model-assisted reflective check-ins triggered by predicted drift risk. The evaluation follows common digital wellbeing field-study practice by reporting usage behavior, self-reported drift, annoyance, usefulness, perceived control, and privacy concern, while adding ML-specific measures such as F1-score, ROC-AUC, first-3-minute performance, expected check-in rate, displayed risk level, and check-in timing feedback. Because empirical data collection is not yet complete, this draft does not report results; all performance values are left as placeholders for later study data. The contribution is a narrow implementation and evaluation framework for predicting and potentially reducing self-reported browser-session drift, not a claim to detect attention, emotion, addiction, or mental health status.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**; *Ubiquitous and mobile computing systems and tools*; • **Computing methodologies** → *Machine learning approaches.*

Additional Key Words and Phrases: digital wellbeing, browser extension, intention alignment, digital drift, self-report, privacy-preserving sensing, session-level prediction, reflective check-ins

1 Introduction

People often open distracting websites for legitimate reasons. A user may visit YouTube to find a tutorial, Reddit to answer a technical question, LinkedIn to check a message, or a news site to follow a specific event. The same sites can also support unplanned passive consumption or avoidance. This ambiguity creates a problem for common digital wellbeing tools: a domain category or elapsed time alone cannot reliably distinguish purposeful use from digital drift.

Existing tools often operationalize undesirable use through screen time, app or website category, or fixed blocking rules [3, 14, 17, 19]. Such mechanisms can be useful for self-control, but they can also generate false positives when a user has a valid reason to visit a monitored website. They may also generate false negatives when a short session is clearly misaligned with intention. Recent work has therefore shifted toward user agency, reflection, friction, and intention alignment. For example, self-nudging systems such as *one sec* introduce a short pause before app access [9]; Purpose Mode reduces distracting interface patterns while preserving access to social media websites [12]; and PauseNow frames digital wellbeing as alignment between intentions and behavior [21].

We use the term *browser-based digital drift* to describe a session-level intention-behavior mismatch: the user begins with one declared purpose but later reports that the visit did not match that intention. This framing is deliberately narrower than addiction, attention detection, or productivity optimization. DriftSense does not infer clinical status or inner mental state. Instead, it collects explicit intention and explicit post-session reflection, then evaluates whether those self-reported labels can be predicted from lightweight browser-session signals.

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DriftSense is designed around a local-first Chrome Manifest V3 extension and a reproducible machine-learning pipeline. In prompted conditions, before accessing a configured domain, the user declares an intended activity using a neutral set of options: *work or study task*, *learning or tutorial*, *specific information*, *communication or community*, *planned entertainment or break*, *open-ended browsing*, or *opened accidentally*. The categories avoid assuming that work is inherently aligned or leisure is inherently drift. During the session, the extension records metadata and aggregate interaction counts, such as duration, click count, scroll count, idle time, tab focus changes, and activity-window summaries. It does not collect page text, screenshots, passwords, private messages, full browsing history, source code, webcam data, or keystroke values. After a configured duration or session boundary, the extension asks whether the visit matched the original intention or became drift. In the model-assisted condition, a locally computed drift-risk probability can trigger a lightweight reflective check-in that displays an estimated risk level and short privacy-safe reason cues, but the extension does not block the website.

The central research questions are:

> *RQ1: Can declared browsing intention combined with lightweight browser activity signals predict self-reported session-level drift better than time-based or domain-based approaches?*
> *RQ2: Does asking users to declare a browsing intention reduce self-reported drift compared with passive monitored browsing?*
> *RQ3: Does a model-assisted reflective check-in reduce self-reported drift beyond a static intention prompt?*

The expected contribution is an implementation and evaluation framework for intention-aware drift prediction and lightweight reflective support in browser sessions. The contribution is not that DriftSense tracks activity, displays a dashboard, or blocks websites. Rather, the contribution is the combination of privacy-preserving activity signals, self-reported drift labels, baseline model comparisons, a three-condition comparison of passive monitoring, static intention prompting, and model-assisted reflection, and an evaluation of whether ML helps time reflective prompts more appropriately than a static prompt alone.

2 Related Work

2.1 Digital Wellbeing and Digital Self-Control Tools

Digital wellbeing research has documented a broad ecosystem of tools that help users manage technology use through blocking, friction, monitoring, goal setting, reminders, interface changes, and reflection [3, 14, 17, 19]. Roffarello and De Russis' systematic review and meta-analysis shows that digital self-control tools vary widely in intervention strategy and evaluation quality, and that many tools emphasize reducing overuse rather than supporting more contextual judgments about why use occurs [17]. Plackett et al.'s journal review of social media use interventions similarly found mixed evidence for limiting-use and abstinence interventions, reinforcing the need to evaluate more contextual approaches rather than assuming that less use is always the appropriate target [19]. Biedermann et al. review digital self-control interventions for distracting media multitasking and identify a mixture of blockers, goal-setting tools, visualizations, and awareness mechanisms [3].

This body of work motivates DriftSense in two ways. First, it shows that time and blocking are common operational levers. Second, it reveals an opportunity for session-level interpretation that treats user intention as a first-class input rather than a background assumption. DriftSense therefore evaluates time and domain baselines, but it does not treat either as sufficient.

2.2 Browser and Social Media Distraction Interventions

Prior HCI systems have explored interventions that change access conditions or interface affordances. Lyngs et al. reviewed digital self-control tools through dual-systems theory and described mechanisms such as commitment, blocking, reminders, and self-monitoring [14]. Later design interventions for social media explored ways to help users maintain self-control without simply removing access [15].

Recent work has focused more directly on platform design patterns. Purpose Mode lets users toggle attention-capturing patterns on social media websites and reports reduced perceived distraction in a field deployment [12]. Research on attention-capture deceptive designs also argues that interface patterns such as infinite scroll and autoplay can shape attentional harms [18]. DriftSense differs from such interface-modification work because it does not remove or alter social media features. Instead, it studies whether session-level drift can be predicted from declared intention and lightweight activity signals, and whether a reflective check-in can help users reassess sessions at risk of drift.

2.3 Intention-Aware and Reflective Interventions

Intention-aware systems directly ask users to articulate goals or create moments of reflection before use. The *one sec* app, evaluated in PNAS, inserts a short breathing pause before opening selected apps and reports reduced target-app opening over time [9]. PauseNow proposes intention-aware recommendations that guide users back to their original goals when they become lost during social media use [21]. StayFocused studies reflective prompts and chatbot support for compulsive smartphone use, finding that reflection can help participants focus longer and resist distractions [13]. MindShift uses large language models to adapt intervention content to reported mental states and contexts in problematic smartphone use interventions [20].

DriftSense borrows the idea that intentions matter, but it uses that idea for measurement, prediction, and lightweight reflective check-ins rather than website blocking or mental-state inference. The pre-session prompt is not meant to block access. The post-session reflection supplies a self-reported session label. This allows DriftSense to evaluate whether intention adds predictive value beyond domain and time, and whether model-triggered reflection can reduce self-reported drift without changing website interfaces.

2.4 Personal Informatics and Activity Tracking

Personal informatics systems help users collect and reflect on data about their behavior. In digital wellbeing, activity summaries and dashboards can make technology use visible, but reflection can be more useful when the data is connected to meaningful goals rather than abstract totals. WellScreen, for example, scaffolds daily reflection by asking people to estimate and report smartphone use, emphasizing self-awareness beyond raw screen time [2]. Work on multi-device digital wellbeing also shows that users' experiences span devices and contexts, complicating simplistic time-based interpretations [16].

DriftSense uses dashboarding and export features only as supporting infrastructure. The dashboard summarizes configured-domain sessions, drift sessions, non-drift sessions, top drift domains, intention distributions, duration, activity patterns, and trends. These summaries help participants and researchers inspect the data, but the paper's central object of evaluation is the drift classifier.

2.5 Regret, Intention Mismatch, and Session-Level Modeling

Regretful or meaningless technology use is closely related to digital drift. Cho et al. studied regretful smartphone use at the app-feature level and showed that regret depends on the kind of activity, not merely the amount of time spent [7]. Guo et al. analyzed a large corpus of smartphone screenshots with multimodal LLMs and found that regret varies by user intention and type of social media use [10]. A recent preprint on regretful social media sessions reports that the gap between intended and actual use predicts regret more strongly than session duration, further supporting intention mismatch as an important unit of analysis [1].

These studies motivate DriftSense's label definition. A drift label is not assigned because a session is long or because the domain is social media. A drift label is assigned when the user explicitly reports that the visit did not match the original intention. This choice accepts that labels are subjective and potentially noisy, but it makes the target construct closer to the experience of drift than a generic time threshold.

2.6 Privacy and Ethical Concerns in Behavioral Tracking

Behavioral tracking creates privacy risks, especially when systems collect fine-grained content, messages, screenshots, or inferred mental states. Digital wellbeing scholarship has argued that tools should respect user agency and avoid framing all technology use as harmful [4, 5, 21]. Digital phenotyping reviews also highlight the need to distinguish behavioral measurement from clinical inference and to avoid overclaiming from sensor data [11].

DriftSense responds by minimizing data collection. The prototype records only monitored domains and metadata needed for session-level modeling. It does not collect page text, passwords, screenshots, messages, full history, source code, keystroke content, webcam data, face identity, or emotion data. The extension stores data locally first and provides export and delete-all-data controls. These constraints reduce model expressiveness, but they keep the prototype aligned with the paper's narrow claim.

3 System Design

3.1 Design Goals

DriftSense is designed for a research setting where participants voluntarily install a browser extension, configure monitored domains, and export a local dataset for analysis. The system has five design goals:

(1) **Condition-aware measurement.** The system should support passive monitoring, static intention prompting, and model-assisted reflective check-ins.
(2) **Lightweight behavioral sensing.** The system should collect aggregate interaction signals that are plausible predictors of drift without collecting page content or sensitive input.
(3) **Local-first privacy.** Data should be stored locally, be exportable by the user, and be deletable through the interface.
(4) **Model-comparable outputs.** The exported schema should support weak baselines, tabular models, prediction at one, three, and five minutes, and condition-level drift-rate comparison.
(5) **Inspectable feedback.** Participants should be able to review aggregate trends, self-reported labels, and model-assisted events without exposing prohibited content or presenting model estimates as facts.

3.2 Chrome Extension Architecture

The planned implementation uses Chrome Manifest V3 with React and TypeScript for the user interface. The extension contains a background service worker for tab/session monitoring, content scripts for aggregate activity counting on

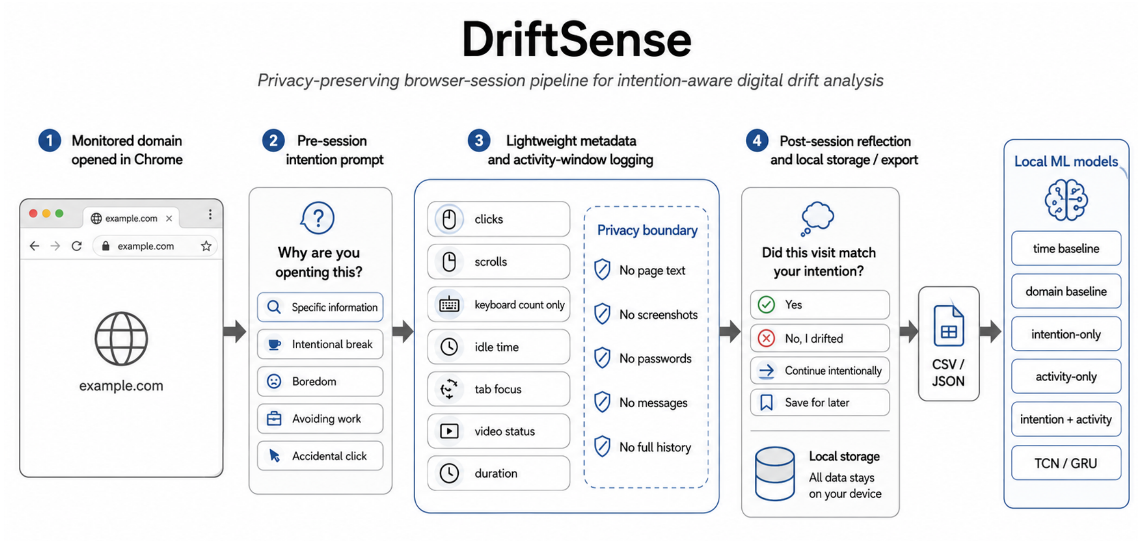

Fig. 1. Planned DriftSense architecture. The extension captures lightweight activity windows and post-session reflection on monitored domains, supports optional intention prompting and model-assisted check-ins, stores data locally, and exports CSV/JSON files for local modeling.

monitored domains, a popup for quick status, an options page for domain configuration and privacy controls, and a dashboard for summaries and export. In the model-assisted condition, the extension computes a local drift-risk probability after an early activity window and can display a short reflective check-in when risk exceeds a tuned threshold.

Figure 1 summarizes the planned DriftSense extension and local machine-learning pipeline.

3.3 Monitored Domains and Study Conditions

Users configure domains that they consider drift-prone or worth studying. Default examples include `youtube.com`, `facebook.com`, `reddit.com`, `instagram.com`, `x.com`, `linkedin.com`, news websites, and custom domains. DriftSense only tracks configured domains.

DriftSense supports three study conditions:

(1) **Passive baseline.** The extension records lightweight metadata on monitored domains and asks a post-session retrospective drift question. There is no pre-session intention prompt.
(2) **Static intention prompt.** The extension asks the user to declare an intention before the monitored-domain session and later asks whether the visit matched that intention.
(3) **Model-assisted prompt.** The extension asks the same pre-session intention question, computes a drift-risk probability after an early activity window, and shows a lightweight reflective check-in if risk exceeds a tuned threshold.

In prompted conditions, when the user opens a monitored domain, the extension displays a pre-session intention prompt:

> *Why are you opening this?*

The response options are:

- Work or study task
- Learning or tutorial
- Specific information
- Communication or community
- Planned entertainment or break
- Open-ended browsing
- Opened accidentally

These choices describe the participant's intended activity rather than labeling an activity as productive or problematic. After selection, the site remains accessible. The prompt is a labeling and reflection mechanism, not a blocker.

In the model-assisted condition, if the local model estimates high drift risk after the first three minutes, DriftSense can show a reflective check-in such as:

> *Drift risk looks high for this session.*
> *Estimated drift risk: 76%.*
> *Possible reasons: longer than intended, high scrolling.*
> *Still here for your original reason?*

Example responses include *Yes, continue*, *I drifted, wrap up*, *Remind me in 5 minutes*, and *Save for later*. The displayed probability is phrased as an estimate rather than a fact. Reason cues are derived only from allowed metadata such as duration, scrolling, idle time, tab switching, or video playback status; they do not use page text, titles, comments, messages, recommendations, or search queries. If pilot participants find exact percentages confusing or stressful, the study interface can show only a coarse risk label such as *low*, *medium*, or *high*, while still exporting the numeric probability for analysis.

3.4 Session Tracking

A session begins when the user opens or focuses a monitored domain. In prompted conditions, the pre-session intention answer is associated with the session. A session ends when the tab is closed, the monitored domain is left, a timeout condition is met, or the user completes the post-session reflection. The system records the following session-level fields:

- `session_id`
- `anonymous_user_id`
- `study_stage`
- `condition`
- `domain`
- `domain_category`
- `declared_intention`
- `start_time` and `end_time`
- `duration_seconds`
- `click_count`
- `scroll_count`
- `keyboard_activity_count`, without key values
- `idle_seconds`

- `active_seconds`
- `tab_focus_loss_count`
- `tab_switch_count`
- `video_playing_seconds`, if detectable
- `checkin_count`
- `model_drift_probability`, for the model-assisted condition
- `model_risk_level`, for the model-assisted condition
- `model_reason_cues`, for the model-assisted condition
- `model_checkin_shown`, for the model-assisted condition
- `model_checkin_response`, for the model-assisted condition
- `model_probability_shown_to_user`, for the model-assisted condition
- `post_session_answer`
- `drift_label`
- `intended_duration_minutes`, if provided
- `actual_duration_seconds`

3.5 Activity Windows

Every 5 or 10 seconds, the content script records an activity-window summary. Each window contains:

- `session_id`
- `timestamp_offset_seconds`
- `clicks_in_window`
- `scroll_events_in_window`
- `keyboard_activity_in_window`, without key values
- `idle_in_window`
- `tab_focused`
- `video_playing`
- `url_domain_only`

These windows support first-3-minute prediction and full-session modeling. They also preserve the privacy boundary: no page text, screenshot pixels, input content, private messages, full browsing history, or source code are collected.

3.6 Post-Session Reflection and Labeling

After a configurable duration or session boundary, DriftSense asks a post-session reflection question. In the static intention and model-assisted conditions, DriftSense asks:

> *Did this visit match your original intention?*

The response options are:

- Yes
- No, I drifted
- Continue intentionally
- Save for later

Open monitored domain → Declare intended activity → Browse session → Post-session reflection

Work/study; learning/tutorial; specific information; communication/community; planned entertainment/break; open-ended browsing; accidental open

Yes, it matched → 0 *No, I drifted* → 1 Other responses → non-binary

Fig. 2. DriftSense session-labeling flow. Prompted conditions ask users to declare an intention before a monitored-domain session and later report whether the visit matched that intention. The passive condition uses a retrospective post-session drift label.

The primary binary labels are:

$$y = \begin{cases} 0, & \text{if answer is } \textit{Yes} \\ 1, & \text{if answer is } \textit{No, I drifted} \end{cases} \tag{1}$$

Continue intentionally and *Save for later* are retained as non-binary outcomes. Depending on the analysis, they may be excluded from binary classification or modeled as separate classes.

In the passive baseline condition, there is no pre-session intention prompt. To avoid changing the user's behavior before browsing, DriftSense instead asks a retrospective post-session question:

> *Did this visit feel intentional, or did it become drift?*

The planned passive-condition labels are *Mostly intentional*, *I drifted*, and *Not sure*. *Mostly intentional* is treated as non-drift, *I drifted* is treated as drift, and *Not sure* is excluded from the main binary outcome or analyzed separately. Because this passive label is retrospective and not identical to the prompted-condition label, condition-level comparisons should be interpreted cautiously.

Figure 2 illustrates the session-labeling flow from monitored-domain access through intention declaration, activity-window logging, and post-session reflection.

3.7 Dashboard and Export

The participant-facing dashboard is a local reflection and transparency interface organized into six views. An *Overview* reports monitored and labeled sessions, self-reported drift rate, average duration, monitoring status, and seven-day and study-to-date summaries. *Trends* visualizes daily and weekly session counts, duration, and drift rate, with optional time-of-day and day-of-week breakdowns. *Intention Alignment* compares intention categories, intended and actual duration where available, domains, and post-session outcomes. These views use descriptive language and do not infer productivity, attention, addiction, or mental state.

The *Model Transparency* view summarizes how many model-assisted check-ins were shown, their estimated probabilities and risk levels, the privacy-safe reason cues displayed, participant responses, and ratings of timing, usefulness, or annoyance. Model estimates are visually separated from self-reported outcomes. The interface states that risk is an estimate used to time reflection, not proof that drift occurred. Exact probabilities can be replaced by low, medium, and high labels if pilot feedback indicates that percentages are confusing or stressful.

The *Session History* view provides a filterable table by date, monitored domain, condition, declared intention, reflection label, and model risk level. A session-detail view shows duration, aggregate activity counts, post-session reflection, and check-in outcome, but never page titles, full URLs, page text, screenshots, messages, or keystroke values. The *Data and Privacy* view allows participants to pause monitoring, configure domains, inspect field definitions, delete an individual session, delete all local data, and export:

- a primary participant CSV named from the anonymous code, such as `P01.csv`;
- an optional activity-windows CSV for early-prediction analysis; and
- an optional complete JSON audit bundle containing allowlisted session and activity-window records.

Participant CSV files share the same 13-column modeling schema and are combined locally into `data.csv`. Missing or non-binary reflection outcomes remain blank in the participant source files and are not silently recoded as non-drift.

All dashboard queries run over locally stored records. The dashboard uses the same structure and availability across Stage 2 conditions and updates after a session or at a delayed aggregate interval rather than acting as an additional real-time intervention. Model details are shown only for records where the model-assisted interface generated them. Missing and uncertain labels remain visible as such and are not silently counted as non-drift. Dashboard preferences are stored locally, and dashboard-opening behavior is not logged unless separately consented to and defined as a secondary measure. The dashboard is included to support reflection, transparency, and data inspection; it is not treated as the paper's novelty claim.

4 Drift Prediction and Model-Assisted Reflection

4.1 Prediction Task

The main task is binary session-level prediction:

> Given declared intention, domain metadata, and lightweight activity signals from a prompted browser session, predict whether the user will later report that the session was drift.

The positive class is the self-reported *No, I drifted* label. The negative class is the self-reported *Yes* label. Sessions labeled *Continue intentionally* or *Save for later* are analyzed separately or excluded from the binary task. The resulting model outputs a session-level drift probability. In the model-assisted condition, this probability is used to decide whether to show a reflective check-in; it is not treated as attention detection.

4.2 Feature Groups

The tabular feature set includes:

- declared intention
- domain category
- time of day
- day of week
- duration so far
- total clicks
- total scrolls
- total keyboard activity count
- idle ratio
- active ratio
- video ratio
- focus loss count
- tab switch count
- average activity per minute
- first-3-minute activity summaries

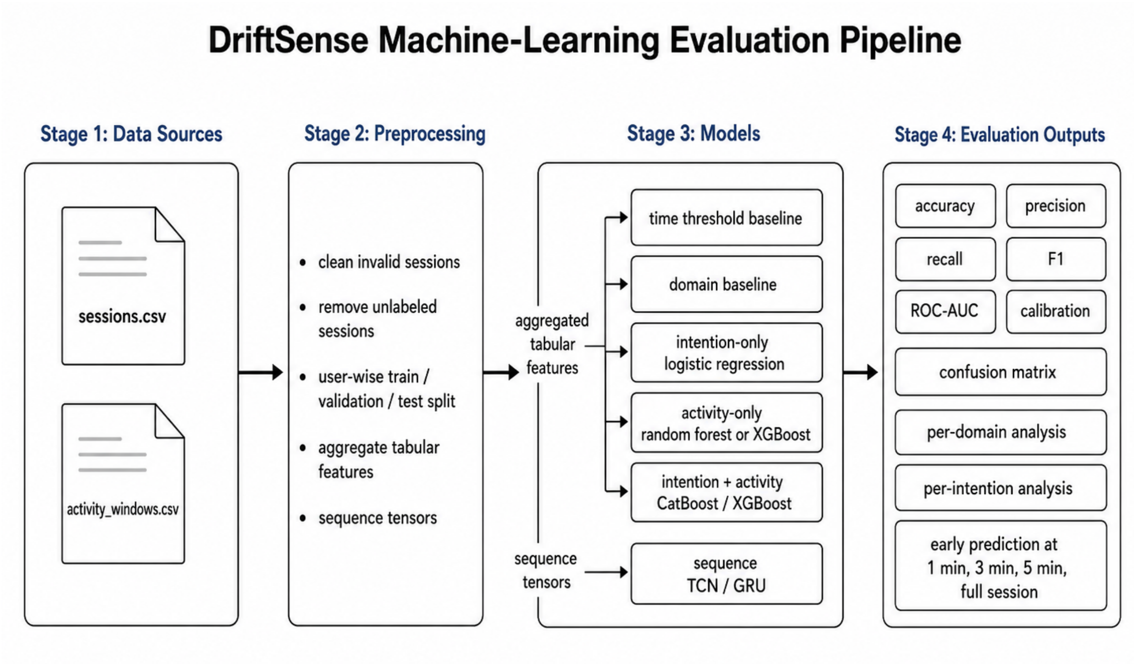

Fig. 3. Planned DriftSense modeling and evaluation pipeline. Exported session and activity-window data are cleaned, split, transformed into first-3-minute and full-session tabular features, evaluated across baselines and intention-aware models, and used to support model-assisted reflective check-ins.

The main feature sets are a first-3-minute feature set for model-assisted check-ins and a full-session feature set for offline evaluation. Activity-window sequences are used to compute aggregate features rather than to train a deep sequence model in the first study.

Figure 3 shows the planned preprocessing, modeling, and evaluation pipeline for tabular drift prediction and model-assisted reflection.

4.3 Baselines

DriftSense evaluates four baselines before the main models:

(1) **Time-threshold baseline.** Predict drift if session duration exceeds a threshold tuned on the validation set.
(2) **Domain baseline.** Predict drift for configured social/video/news domains marked as drift-prone.
(3) **Intention-only logistic regression.** Predict from declared intention and basic context features only.
(4) **Activity-only model.** Predict from aggregate activity signals without declared intention.

These baselines test whether the proposed intention-plus-activity models add value beyond common digital wellbeing heuristics.

4.4 Main Models

The primary tabular model is Random Forest, CatBoost, or XGBoost using intention, domain, time, and aggregate activity features [6, 8]. These models are appropriate for heterogeneous tabular features and can provide feature importance summaries. Random Forest is acceptable for the first field study if CatBoost or XGBoost integration is too heavy.

Deep sequence models are not part of the main planned study unless the dataset becomes substantially larger. This keeps the evaluation focused on whether declared intention and lightweight activity metadata add value beyond simple baselines.

4.5 Early Prediction

Early prediction evaluates whether drift can be estimated before the full session ends. The simplified study evaluates two feature windows:

- first 3 minutes
- full session

The first-3-minute model is used for the model-assisted condition. If predicted drift probability exceeds a threshold tuned on the training data, DriftSense shows a short reflective check-in. The check-in is intended to support user reassessment, not to block access.

5 Methodology

5.1 Study Design

The planned study has two stages. Stage 1 collects an intention-prompt dataset for training and validating a drift-risk model. Stage 2 freezes the trained model and compares three field-study conditions: passive monitoring, static intention prompting, and model-assisted reflective check-ins. Participants install DriftSense, configure monitored domains, browse normally, and export the locally stored dataset at the end of the study.

Because real data collection has not yet been completed, all sample sizes and results are placeholders:

- Participants: [INSERT PARTICIPANT COUNT]
- Stage 1 duration: [INSERT DURATION]
- Stage 2 duration: [INSERT DURATION]
- Total monitored-domain sessions: [INSERT SESSION COUNT]
- Labeled sessions: [INSERT LABELED SESSION COUNT]
- Drift class proportion: [INSERT CLASS DISTRIBUTION]

5.2 Stage 1: Model-Training Dataset

Stage 1 uses the static intention-prompt condition only. Participants declare why they are opening a monitored domain, browse normally, and answer the post-session reflection question. No model-assisted check-ins are shown in this stage. The recommended Stage 1 target is 10–15 participants over 4–5 days, with approximately 250–500 labeled sessions if participant activity is sufficient.

The Stage 1 data is used to train and validate the drift-risk model. The main model uses declared intention, domain category, time features, and first-3-minute activity features. A validation set is used to choose a conservative check-in threshold for Stage 2, for example a threshold that prioritizes avoiding excessive false nudges.

5.3 Stage 2: Three-Condition Field Evaluation

Stage 2 compares three conditions:

(1) **Passive baseline.** DriftSense records lightweight metadata on monitored domains and asks only a post-session retrospective drift question. This condition approximates normal monitored browsing without a pre-session intention prompt.
(2) **Static intention prompt.** DriftSense asks the pre-session intention question and later asks whether the session matched the original intention. No model-assisted check-in is shown.
(3) **Model-assisted prompt.** DriftSense asks the same pre-session intention question, computes drift probability after the first three minutes, and shows a reflective check-in if the probability exceeds the tuned threshold.

The recommended Stage 2 target is 20–25 participants over 7 days, with approximately 400–800 labeled sessions. If implementation time permits, condition order should be counterbalanced across participants. If counterbalancing is not feasible, a fixed schedule may be used, such as two days of passive baseline, two days of static intention prompting, and three days of model-assisted prompting. A fixed order must be reported as a limitation because learning or novelty effects may influence condition differences.

5.4 Participants

Participants should be adults who use Chrome or Chromium-based browsers and are comfortable installing a research extension. Recruitment may focus on students or knowledge workers because browser-based drift is common in study and work settings. The study should avoid recruiting participants on the basis of mental health status, addiction, ADHD, or other clinical categories because DriftSense is not designed for diagnosis or treatment.

5.5 Consent and Privacy Procedure

Before installation, participants receive a consent form explaining what DriftSense collects and what it does not collect. The consent process should state that the extension records only configured-domain metadata, condition assignment, declared intention where applicable, aggregate interaction counts, activity-window summaries, model drift probability where applicable, displayed risk level and reason cues where applicable, check-in response where applicable, whether probability was shown, and post-session reflections. It should explicitly state that DriftSense does not collect page text, passwords, screenshots, private messages, full browsing history, source code, keystroke values, webcam data, face identity, or emotion data.

Participants should be able to pause monitoring, remove monitored domains, export their data, and delete all locally stored data. Exported datasets should use anonymous participant IDs. Any real dataset used for publication should be reviewed for accidental sensitive fields before analysis.

5.6 Procedure

The planned procedure has nine steps:

(1) Participants complete consent and a short onboarding survey about browser use and monitored domains.
(2) Participants install DriftSense and configure default or custom domains.
(3) Stage 1 participants use the static intention-prompt condition to generate training data.
(4) The research team trains and freezes the drift-risk model using Stage 1 data.
(5) Stage 2 participants browse under passive, static prompt, and model-assisted conditions.

(6) DriftSense records lightweight session metadata and activity windows while monitored domains are active.
(7) Participants answer post-session reflection prompts appropriate to the condition.
(8) Participants can inspect delayed aggregate summaries and their own session records through the condition-neutral local dashboard.
(9) At the end of the study, participants export the dataset and complete a post-study survey; optional short interviews collect qualitative feedback about usefulness, burden, dashboard comprehension, and privacy concerns.

5.7 Measures

The evaluation is designed to remain comparable to prior digital wellbeing and self-control studies while adding an ML-specific layer. Like related field studies, DriftSense reports usage behavior and self-reported experience: session counts, duration, drift-session percentage, prompt usefulness, annoyance, perceived control, and privacy concern. The additional contribution is to evaluate whether a learned model can time reflective check-ins more appropriately than a static prompt.

The prediction evaluation reports:

- accuracy
- precision
- recall
- F1-score
- ROC-AUC
- confusion matrix
- first-3-minute versus full-session performance
- expected check-in rate at the selected threshold

The condition-level evaluation reports:

- drift-session percentage by condition
- session count by condition
- mean and median session duration by condition
- model check-ins shown
- percentage of sessions where a model check-in was shown
- average model drift probability when a check-in was shown
- displayed model risk level
- displayed model reason cues
- check-in response distribution
- prompt annoyance
- perceived control
- perceived usefulness
- privacy concern
- participant feedback on whether model-assisted check-ins appeared at appropriate times
- dashboard comprehension, perceived usefulness, and trust in the distinction between self-reported labels and model estimates

These user-study measures are descriptive and should not be used to claim clinical benefit. The main behavioral outcome is self-reported drift rate, not objectively detected distraction. The model-specific outcome is whether the predicted risk helped trigger a timely reflective prompt without producing excessive false or annoying nudges.

5.8 Data Processing

The analysis pipeline validates participant files such as `P01.csv`, combines complete binary-labeled rows into `data.csv`, and optionally loads the activity-windows CSV for early-prediction analysis. It removes invalid sessions and separates Stage 1 model-training data from Stage 2 condition-evaluation data. For prediction modeling, participant-wise splits are preferred to reduce leakage from the same participant's repeated browsing patterns. When participant-wise splitting is not possible because the dataset is too small, the paper should report this limitation clearly.

Categorical variables are encoded using one-hot encoding or model-native categorical handling. Numeric activity features are standardized for models that require scaling. First-3-minute features are generated for model-assisted check-ins, while full-session features are generated for offline comparison.

5.9 Analysis Plan

The prediction hypothesis is that models using both intention and activity will outperform time-only and domain-only baselines on F1-score and ROC-AUC. Secondary prediction analyses compare intention-only versus activity-only models and inspect feature importance for the strongest tabular model. For the model-assisted condition, the selected first-3-minute model is also evaluated by expected check-in rate and threshold behavior, because a highly sensitive model may be statistically interesting but unusable if it prompts too often.

The condition-level hypothesis is that the model-assisted prompt condition will have lower self-reported drift than the static intention-prompt condition, and that the static intention-prompt condition will have lower self-reported drift than the passive baseline. Because the passive baseline uses a retrospective drift question rather than a pre-session intention prompt, this comparison should be framed cautiously as evidence about self-reported drift rates under different interaction designs.

The ML-assistance analysis asks whether model-triggered reflection feels timely and useful. This analysis reports how often check-ins are shown, which risk levels and reason cues are displayed, how participants respond, and whether participants report that the check-ins appeared at appropriate moments. The paper should avoid claiming that a probability proves that a user was drifting; it should instead describe the probability as an estimate used to decide whether a reflective prompt may be worth showing.

No results should be fabricated. If the dataset is incomplete, tables should use placeholders such as [INSERT RESULT]. If the dataset is small, the paper should emphasize uncertainty, report confidence intervals or bootstrap intervals where possible, and avoid generalizing beyond the study sample.

References

[1] Sally Ahmed, Jan Enkmann, Kye Shimizu, Ivy Yip, Vincent Beermann, Ayse Alomar, Falk Uebernickel, and Pattie Maes. 2026. Before You Scroll Again: Predicting Regretful Social Media Sessions from In-the-Wild Contextual and Wearable Sensing. Preprint. arXiv:2606.08965 [cs.HC] doi:10.48550/arXiv.2606.08965

[2] Ananya Bhat, Joshua Shi, Sean A. Munson, and Alexis Hiniker. 2026. "In My Defense, Only Three Hours on Instagram": Designing Toward Digital Self-Awareness and Wellbeing. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*. ACM. Forthcoming/early online. doi:10.1145/3772318.3790263

[3] Daniel Biedermann, Jan Schneider, and Hendrik Drachsler. 2021. Digital Self-Control Interventions for Distracting Media Multitasking: A Systematic Review. *Journal of Computer Assisted Learning* 37, 5 (2021), 1217–1231. doi:10.1111/jcal.12581

[4] Moritz Büchi. 2021. Digital Well-Being Theory and Research. *New Media & Society* (2021). doi:10.1177/14614448211056851

[5] Christopher Burr, Mariarosaria Taddeo, and Luciano Floridi. 2020. The Ethics of Digital Well-Being: A Thematic Review. *Science and Engineering Ethics* 26, 4 (2020), 2313–2343. doi:10.1007/s11948-020-00175-8

[6] Tianqi Chen and Carlos Guestrin. 2016. XGBoost: A Scalable Tree Boosting System. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. ACM, 785–794. doi:10.1145/2939672.2939785

[7] Hyunsung Cho, Daeun Choi, Donghwi Kim, Wan Ju Kang, Eun Kyoung Choe, and Sung-Ju Lee. 2021. Reflect, Not Regret: Understanding Regretful Smartphone Use with App Feature-Level Analysis. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW2 (2021), Article 456. doi:10.1145/3479600

[8] Anna Veronika Dorogush, Vasily Ershov, and Andrey Gulin. 2018. CatBoost: Gradient Boosting with Categorical Features Support. In *Workshop on ML Systems at NeurIPS*. arXiv:1810.11363

[9] David J. Grüning, Frederik Riedel, and Philipp Lorenz-Spreen. 2023. Directing Smartphone Use Through the Self-Nudge App One Sec. *Proceedings of the National Academy of Sciences of the United States of America* 120, 8 (2023), e2213114120. doi:10.1073/pnas.2213114120

[10] Longjie Guo, Yue Fu, Xiran Lin, Xuhai Xu, Yung-Ju Chang, and Alexis Hiniker. 2025. What Social Media Use Do People Regret? An Analysis of 34K Smartphone Screenshots with Multimodal LLM. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. ACM, 1–23. doi:10.1145/3706598.3713724

[11] Gustavo Lazarotto Schroeder, Rosemary Francisco, and Jorge Luis Victória Barbosa. 2026. Digital Phenotyping for Problematic Technology Use: A Systematic Literature Review and Taxonomy. *Current Addiction Reports* 13, 1 (2026). doi:10.1007/s40429-026-00759-7

[12] Hao-Ping (Hank) Lee, Yi-Shyuan Chiang, Lan Gao, Stephanie Yang, Philipp Winter, and Sauvik Das. 2025. Purpose Mode: Reducing Distraction Through Toggling Attention Capture Damaging Patterns on Social Media Web Sites. *ACM Transactions on Computer-Human Interaction* 32, 1 (2025), 1–41. doi:10.1145/3711841

[13] Zhuoyang Li, Minhui Liang, Ray Lc, and Yuhan Luo. 2024. StayFocused: Examining the Effects of Reflective Prompts and Chatbot Support on Compulsive Smartphone Use. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. ACM, 1–19. doi:10.1145/3613904.3642479

[14] Ulrik Lyngs, Kai Lukoff, Petr Slovak, Reuben Binns, Adam Slack, Michael Inzlicht, Max Van Kleek, and Nigel Shadbolt. 2019. Self-Control in Cyberspace: Applying Dual Systems Theory to a Review of Digital Self-Control Tools. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. ACM, 1–18. doi:10.1145/3290605.3300361

[15] Ulrik Lyngs, Kai Lukoff, Petr Slovak, William Seymour, Helena Webb, Marina Jirotka, Jun Zhao, Max Van Kleek, and Nigel Shadbolt. 2020. "I Just Want to Hack Myself to Not Get Distracted": Evaluating Design Interventions for Self-Control on Facebook. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. ACM, 1–15. doi:10.1145/3313831.3376672

[16] Alberto Monge Roffarello and Luigi De Russis. 2021. Coping with Digital Wellbeing in a Multi-Device World. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. ACM, 1–14. doi:10.1145/3411764.3445076

[17] Alberto Monge Roffarello and Luigi De Russis. 2023. Achieving Digital Wellbeing Through Digital Self-Control Tools: A Systematic Review and Meta-Analysis. *ACM Transactions on Computer-Human Interaction* 30, 4 (2023), 1–66. doi:10.1145/3571810

[18] Alberto Monge Roffarello, Kai Lukoff, and Luigi De Russis. 2023. Defining and Identifying Attention Capture Deceptive Designs in Digital Interfaces. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. ACM, 1–19. doi:10.1145/3544548.3580729

[19] Ruth Plackett, Alexandra Blyth, and Patricia Schartau. 2023. The Impact of Social Media Use Interventions on Mental Well-Being: Systematic Review. *Journal of Medical Internet Research* 25 (2023), e44922. doi:10.2196/44922

[20] Ruolan Wu, Chun Yu, Xiaole Pan, Yujia Liu, Ningning Zhang, Yue Fu, Yuhan Wang, Zhi Zheng, Li Chen, Qiaolei Jiang, Xuhai Xu, and Yuanchun Shi. 2024. MindShift: Leveraging Large Language Models for Mental-States-Based Problematic Smartphone Use Intervention. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. ACM, 1–24. doi:10.1145/3613904.3642790

[21] Yixue Zhao, Tianyi Li, and Michael Sobolev. 2024. Digital Wellbeing Redefined: Toward User-Centric Approach for Positive Social Media Engagement. In *Proceedings of the IEEE/ACM 11th International Conference on Mobile Software Engineering and Systems*. ACM, 1–5. doi:10.1145/3647632.3651392